

EXCEL GUIDE

Basic Terms in Excel

There are two basic ways to perform calculations in Excel: Formulas and Functions.

1. Formulas

In Excel, a formula is an expression that operates on values in a range of cells or a cell. For example, `=A1+A2+A3`, which finds the sum of the range of values from cell A1 to cell A3.

2. Functions

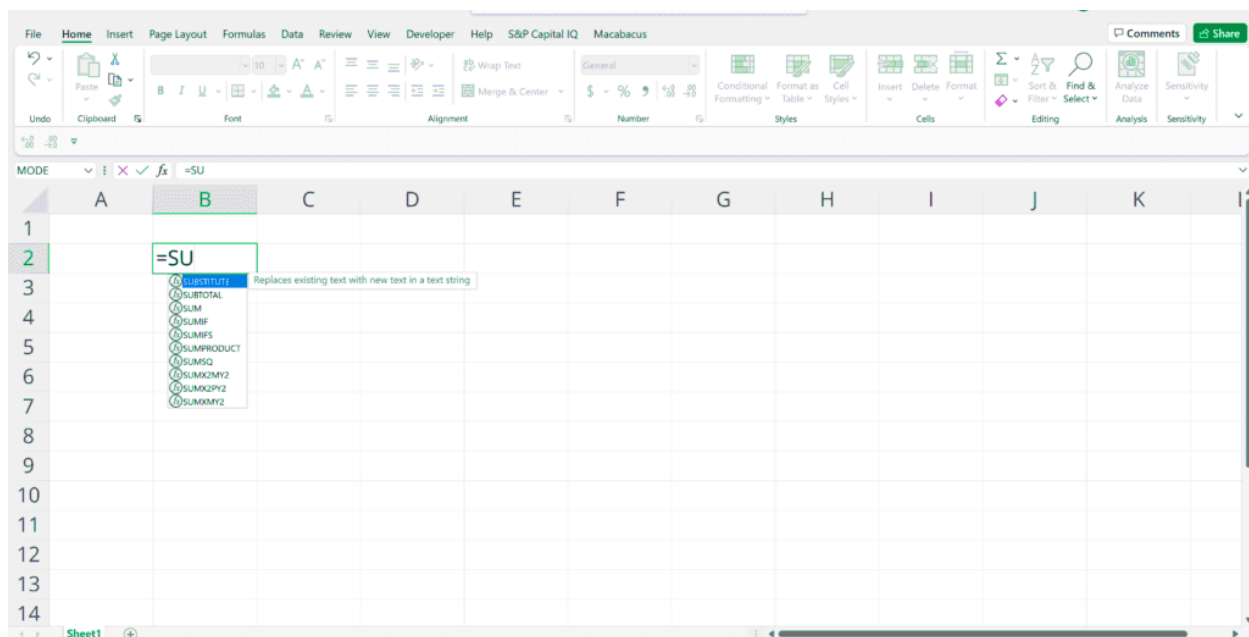
Functions are predefined formulas in Excel. They eliminate laborious manual entry of formulas while giving them human-friendly names. For example: `=SUM(A1:A3)`. The function sums all the values from A1 to A3.

Five Time-saving Ways to Insert Data into Excel

1. Simple insertion: Typing a formula inside the cell

Typing a formula in a cell or the formula bar is the most straightforward method of inserting basic Excel formulas. The process usually starts by typing an equal sign, followed by the name of an Excel function.

when you start typing the name of the function, a pop-up function hint will show (see below). It's from this list you'll select your preference. However, don't press the Enter key after making your selection. Instead, press the Tab key and Excel will automatically fill in the function name.



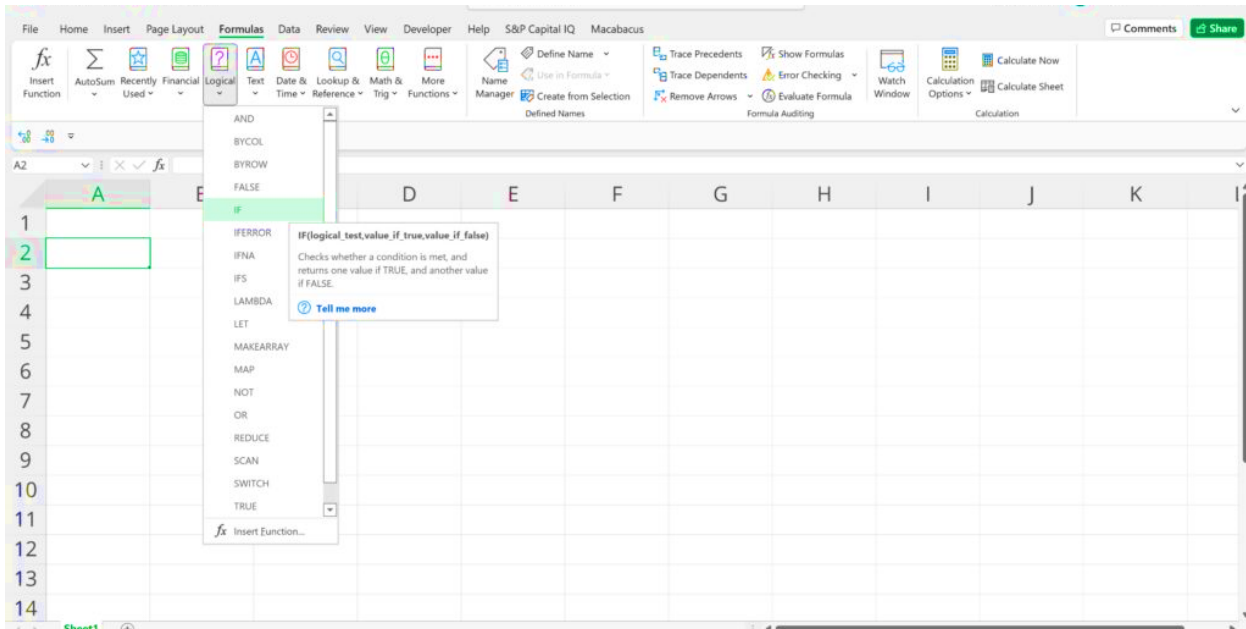
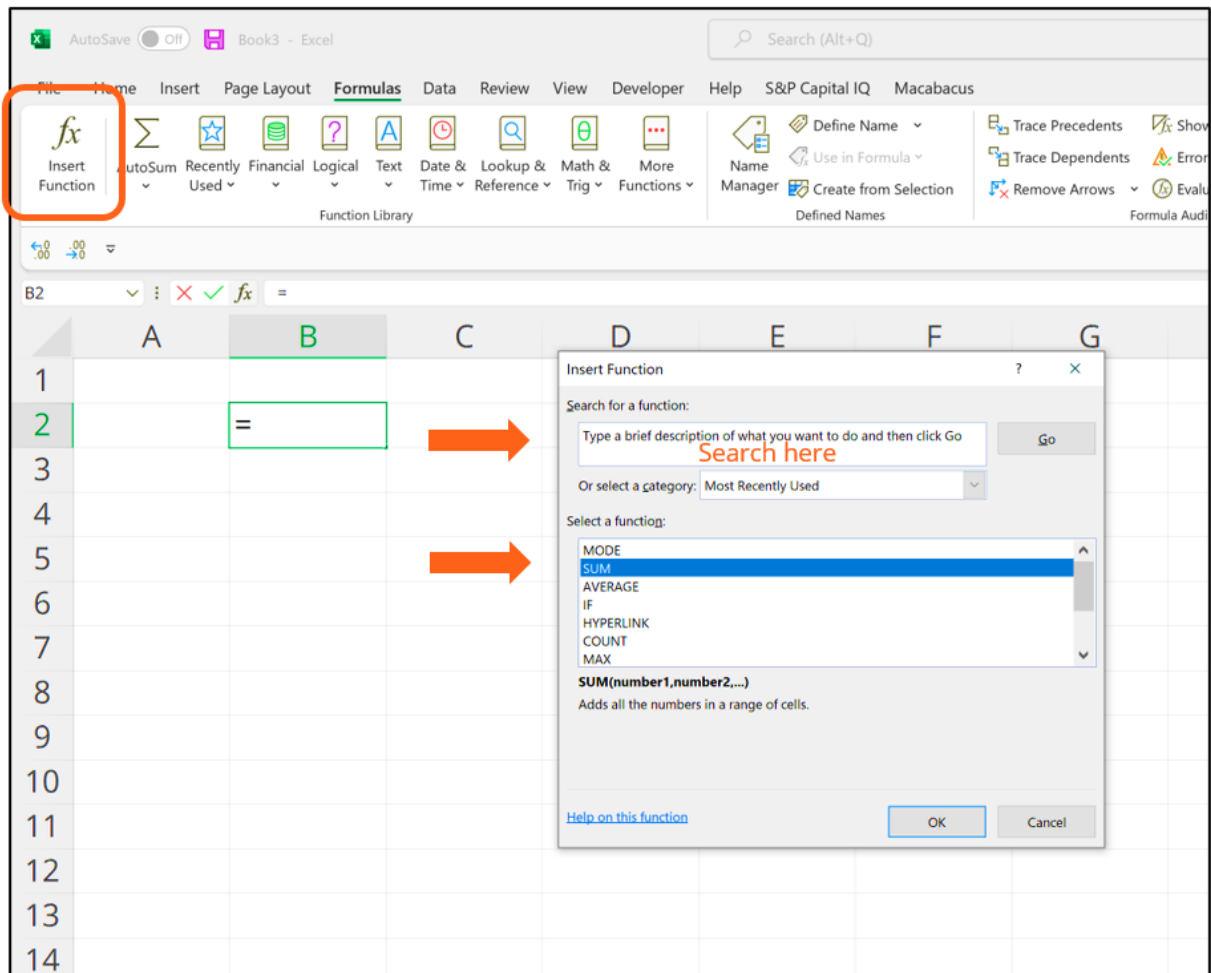
2. Using Insert Function Option from Formulas Tab

If you want full control of your function's insertion, using the Excel Insert Function dialogue box is all you ever need. To achieve this, go to the Formulas tab and select the first menu labeled Insert Function. The dialogue box will contain all the functions you need to complete your financial analysis.

3. Selecting a Formula from One of the Groups in Formula Tab

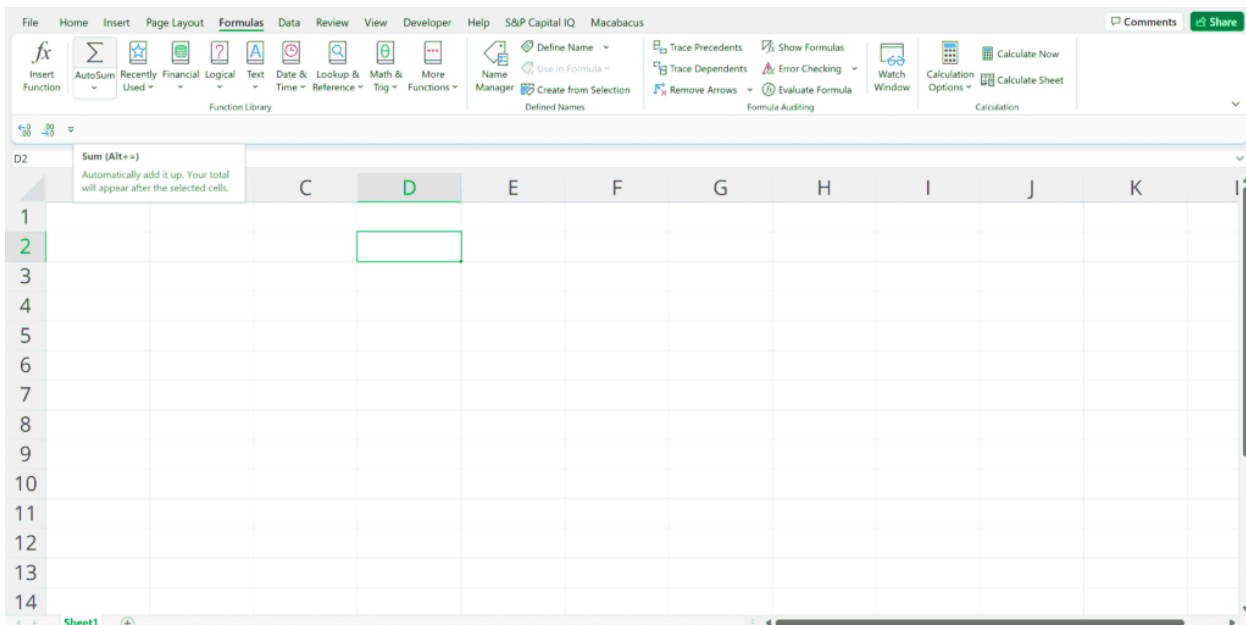
This option is for those who want to use mostly used functions quickly. To find this menu, navigate to the Formulas tab and select your preferred group. Click to show a sub-menu filled with a list of functions.

From there, you can select your preference. However, if you find your preferred group is not on the tab, click on the More Functions option – it's probably just hidden there.



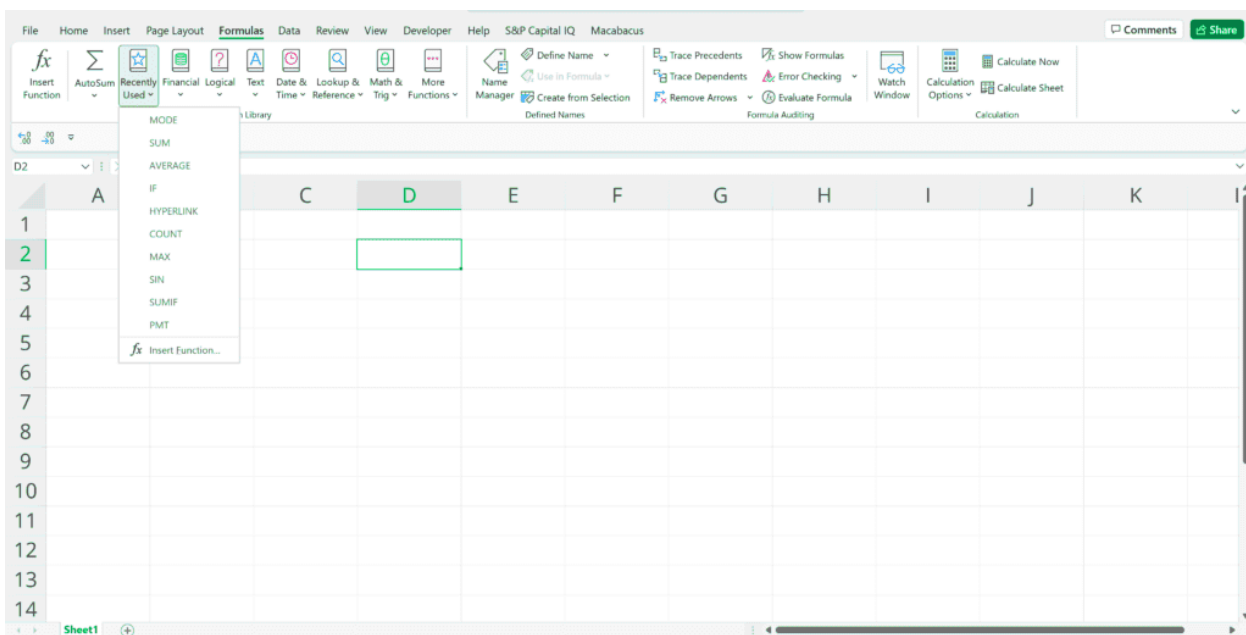
4. Using AutoSum Option

For quick and everyday tasks, the AutoSum function is your go-to option. Navigate to the **Formulas** tab and click the AutoSum option. Then click the caret to show other hidden formulas. This option is also available in the Home tab.



5. Quick Insert: Use Recently Used Tabs

If you find re-typing your most recent formula a monotonous task, then use the Recently Used selection. It's on the Formulas tab, a third menu option just next to AutoSum.



1. SUM

The SUM function is the first must-know formula in Excel. It usually aggregates values from a selection of columns or rows from your selected range.

=SUM(*number1*, [*number2*], ...)

Example:

=SUM(B2:G2) – A simple selection that sums the values of a row.

=SUM(A2:A8) – A simple selection that sums the values of a column.

=SUM(A2:A7, A9, A12:A15) – A sophisticated collection that sums values from range A2 to A7, skips A8, adds A9, jumps A10 and A11, then finally adds from A12 to A15.

=SUM(A2:A8)/20 – Shows you can also turn your function into a formula.

	A	B	C	D	E	F	G	H	I	J
1	Country	Population								
2	China	1,389,618,778								
3	India	1,311,559,204								
4	USA	331,883,986								
5	Indonesia	264,935,824								
6	Pakistan	210,797,836								
7	Brazil	210,301,591								
8	Nigeria	208,679,114								
9	Bangladesh	161,062,905								
10	Russia	141,944,641								
11	Mexico	127,318,112								
12	Total	=SUM(B2:B11)	Output = 4,358,101,991							
13										
14										
15										

2. AVERAGE

The AVERAGE function should remind you of simple averages of data, such as the average number of shareholders in a given shareholding pool.

=AVERAGE(number1, [number2], ...)

Example:

=AVERAGE(B2:B11) – Shows a simple average, also similar to (SUM(B2:B11)/10)

	A	B	C	D	E	F	G	H	I
1	Country	Population							
2	China	1,389,618,778							
3	India	1,311,559,204							
4	USA	331,883,986							
5	Indonesia	264,935,824							
6	Pakistan	210,797,836							
7	Brazil	210,301,591							
8	Nigeria	208,679,114							
9	Bangladesh	161,062,905							
10	Russia	141,944,641							
11	Mexico	127,318,112							
12	Total	=AVERAGE(B2:B11)	Output = 435,810,199						
13									
14									
15									

3. COUNT

The COUNT function counts all cells in a given range that contain only numeric values.

=COUNT(value1, [value2], ...)

Example:

COUNT(A:A) – Counts all values that are numerical in A column. However, you must adjust the range inside the formula to count rows.

COUNT(A1:C1) – Now it can count rows.

	A	B	C	D	E	F	G	H	I
1	Country	Population							
2	China	1,389,618,778							
3	India	1,311,559,204							
4	USA	331,883,986							
5	Indonesia	264,935,824							
6	Pakistan	210,797,836							
7		Empty	Skips non-numerical values						
8	Brazil	210,301,591							
9	Nigeria	208,679,114							
10			Skips empty cells						
11	Bangladesh	161,062,905							
12	Russia	141,944,641							
13	Mexico	127,318,112							
14	Total	=COUNT(B2:B13)	Output = 10						
15									

4. COUNTA

Like the COUNT function, COUNTA counts all cells in a given range. However, it counts all cells regardless of type. That is, unlike COUNT that only counts numerics, it also counts dates, times, strings, logical values, errors, empty string, or text.

=COUNTA(value1, [value2], ...)

Example:

COUNTA(C2:C13) – Counts rows 2 to 13 in column C regardless of type. However, like COUNT, you can't use the same formula to count rows. You must make an adjustment to the selection inside the brackets – for example, **COUNTA(C2:H2)** will count columns C to H

	A	B	C	D	E	F	G	H	I
1	Country	Population							
2	China	1,389,618,778							
3	India	1,311,559,204							
4	USA	331,883,986							
5	Indonesia	264,935,824							
6	Pakistan	210,797,836							
7		Empty	Counts all entries						
8	Brazil	210,301,591							
9	Nigeria	208,679,114							
10			ONLY skips empty cells						
11	Bangladesh	161,062,905							
12	Russia	141,944,641							
13	Mexico	127,318,112							
14	Total	=COUNTA(B2:B13)	Output = 11						
15									

5. IF

The IF function is often used when you want to sort your data according to a given logic. The best part of the IF formula is that you can embed formulas and functions in it.

Example:

=IF(SUM(C1:C10) > SUM(D1:D10), SUM(C1:C10), SUM(D1:D10)) – An example of a complex IF statement. First, it sums C1 to C10 and D1 to D10, then it compares the sum. If the sum of C1 to C10 is greater than the sum of D1 to D10, then it makes the value of a cell equal to the sum of C1 to C10.

6. TRIM

=TRIM(*text*) Example:

File Home Insert Page Layout Formulas Data Review View Developer Help S&P Capital IQ Macabacus

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10R x 1C =TRIM(A2)

	A	B	C	D	E	F	G	H	I
1	Raw Data	Trim							
2	China 2019	China 2019							
3	2020 India	2020 India							
4	USA 2021	USA 2021							
5	Indonesia 2022	Indonesia 2022							
6	2023 Pakistan	2023 Pakistan							
7	Brazil 2024	Brazil 2024							
8	Nigeria is a country	Nigeria is a country							
9	Bangladesh 4444	Bangladesh 4444							
10	Russia 123456	Russia 123456							
11	2025 Mexico	2025 Mexico							
12									
13									
14									
15									

Sheet1 Sheet1 (2) Sheet1 (3)

7. MAX & MIN

The MAX and MIN functions help in finding the maximum number and the minimum number in a range of values.

=MIN(*number1*, [number2], ...)

Example:

=MIN(B2:C11) – Finds the minimum number between column B from B2 and column C from C2 to row 11 in both columns B and C.

=MAX(*number1*, [number2], ...)

Example:

=MAX(B2:C11) – Similarly, it finds the maximum number between column B from B2 and column C from C2 to row 11 in both columns B and C.

[illegible]

AutoSave OFF Exhibits for Basic Excel Formulas.xlsx Jeffrey Schmidt

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MODE fx =MIN(B2:B11)

	A	B	C	D	E	F	G	H	I	J
1	Country	Population								
2	China	1,389,618,778								
3	India	1,311,559,204								
4	USA	331,883,986								
5	Indonesia	264,935,824								
6	Pakistan	210,797,836								
7	Brazil	210,301,591								
8	Nigeria	208,679,114								
9	Bangladesh	161,062,905								
10	Russia	141,944,641								
11	Mexico	127,318,112								
12	MIN	=MIN(B2:B11)	127,318,112							
13										
14										
15										

Sheet1 Sheet1 (2) Sheet1 (3) Sheet1 (4) Sheet1 (5)

What is VLOOKUP in Excel?

VLOOKUP stands for Vertical Lookup. As the name specifies, VLOOKUP is a built-in Excel function that helps you look for a specified value by searching for it vertically across the sheet. VLOOKUP in Excel may sound complicated, but you will find out that it is a very easy and useful tool once you try it. Look at the example below to understand VLOOKUP.

The VLOOKUP formula below looks for a Company name with Company ID 3.

The screenshot shows an Excel spreadsheet with a data table and a VLOOKUP formula. The formula bar at the top displays `=VLOOKUP(H9,A6:F15,2,FALSE)`. The data table has columns: ID, Company, Product, TypeName, Ram, and Price_euros. Row 9 contains the lookup value 3 in the ID column. To the right, a separate table shows the result: Company ID 3 corresponds to Company HP. Red arrows and numbers 1-6 label the formula components: 1 points to the lookup value (H9), 2 points to the table array (A6:F15), 3 points to the column index (2), and 4 points to the match type (FALSE). A red arrow labeled 'Lookup value' points to the value 3 in the lookup table, and a red arrow labeled 'Result' points to the value HP in the result table.

ID	Company	Product	TypeName	Ram	Price_euros
1	Apple	MacBook Pro	Ultrabook	8	1339.69
2	Lenovo	IdeaPad 320-15ISK	Notebook	4	369
3	HP	ProBook 440	Notebook	8	860
4	Microsoft	Surface Laptop	Ultrabook	4	1089
5	Dell	Inspiron 3567	Notebook	4	498.9
6	Acer	Aspire 3	Notebook	4	400
7	Apple	Macbook Air	Ultrabook	8	1158.7
8	Asus	ZenBook	Ultrabook	16	1495
9	Acer	Swift 3	Ultrabook	8	770
10	HP	Pavilion 15	Ultrabook	8	699

Company ID	Company
3	HP

How To Find an Exact Match Using VLOOKUP?

VLOOKUP makes it effortless to look for an exact match from the table. Let's take a look at how to do this with the help of an example:

- In the example below, we are using the VLOOKUP function to find the value of the exact match of ID from the given table. So, we set the first parameter as the lookup value, which is the cell H5.

The screenshot shows an Excel spreadsheet with a data table and a VLOOKUP formula. The data table has columns: ID, Company, Product, TypeName, Ram, and Price_euros. Row 5 contains the lookup value 4 in the ID column. To the right, a separate table shows the result: ID 4 corresponds to Product Surface Laptop. The formula bar at the top displays `=VLOOKUP(H5,A2:F11,3,FALSE)`. Red arrows and numbers 1-4 label the formula components: 1 points to the lookup value (H5), 2 points to the table array (A2:F11), 3 points to the column index (3), and 4 points to the match type (FALSE).

ID	Company	Product	TypeName	Ram	Price_euros
1	Apple	MacBook Pro	Ultrabook	8GB	1339.69
2	Lenovo	IdeaPad 320-15ISK	Notebook	4GB	369
3	HP	ProBook 440	Notebook	8GB	860
4	Microsoft	Surface Laptop	Ultrabook	4GB	1089
5	Dell	Inspiron 3567	Notebook	4GB	498.9
6	Acer	Aspire 3	Notebook	4GB	400
7	Apple	Macbook Air	Ultrabook	8GB	1158.7
8	Asus	ZenBook	Ultrabook	16GB	1495
9	Acer	Swift 3	Ultrabook	8GB	770
10	HP	Pavilion 15	Ultrabook	8GB	699

ID	Product
4	Surface Laptop

- We specify the location of the table in the second argument. As you can see, the table location is A2:F11.
- The third argument specifies the Column Index number. This tells us what value should be returned from the row that we are looking up for. In the example, the product column is 3.
- The last argument is a Boolean Expression. Here, the value is set to FALSE for the VLOOKUP function to return an exact match for the value. An N/A error is displayed in case the exact value is not found.

	A	B	C	D	E	F	G	H	I	J
1	ID	Company	Product	TypeName	Ram	Price_euros				
2	1	Apple	MacBook Pro	Ultrabook	8GB	1339.69				
3	2	Lenovo	IdeaPad 320-15ISK	Notebook	4GB	369				
4	3	HP	ProBook 440	Notebook	8GB	860		ID	Product	
5	4	Microsoft	Surface Laptop	Ultrabook	4GB	1089		4	Surface Laptop	
6	5	Dell	Inspiron 3567	Notebook	4GB	498.9				
7	6	Acer	Aspire 3	Notebook	4GB	400				
8	7	Apple	Macbook Air	Ultrabook	8GB	1158.7				
9	8	Asus	ZenBook	Ultrabook	16GB	1495				
10	9	Acer	Swift 3	Ultrabook	8GB	770				
11	10	HP	Pavilion 15	Ultrabook	8GB	699				
12										

With the help of VLOOKUP in Excel, we can look for an approximate match as well. You will learn this in the next section.

How To Find Approximate Match Using VLOOKUP?

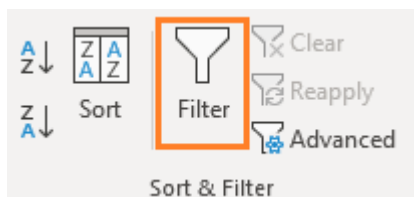
Approximate Match works by finding the next largest value that is lesser than the lookup value, which we specify.

In the example below, we use the VLOOKUP function to find out how much RAM specification a laptop priced 1300 EUR has. Also, we know this value is not present in the table. So, let's use the Approximate Match to find the solution. We need to sort the first column in ascending order. If not, VLOOKUP will return incorrect values.

- First, copy the price and RAM details to a new location, and specify your lookup value. Here, the lookup value is \$1300.

	A	B	C	D	E
1	Price_euros	Ram			
2	369	4			
3	400	4			
4	498.9	4		PRICE Lookup	RAM (GB)
5	699	8		1300	
6	770	8			
7	860	8			
8	1158.7	8			
9	1339.69	8			
10	1495	16			
11					

Next, select your data range and click on the filter option to sort the values of the first column based on ascending order. After you click on the filter option, you will see the filter buttons enabled on your column headers.



- Filter the price details in ascending order. Click on OK.

1 Price_euros Ram

A Z Sort Smallest to Largest

Z A Sort Largest to Smallest

Sort by Color >

Sheet View >

Clear Filter From "Price_euros"

Filter by Color >

Number Filters >

Search

☒ (Select All)

☒ 369

☒ 400

☒ 498.9

☒ 699

☒ 770

☒ 860

☒ 1158.7

☒ 1339.69

OK Cancel

- Now, enter your VLOOKUP formula. The first argument will be the VLOOKUP value. The second argument specifies the table range. In the third argument, we give the column number, so that values for that column are returned. Finally, the last argument is set to TRUE. This will allow the VLOOKUP function to find an approximate match for the value.

SUM				=VLOOKUP(D5,A1:B10,2,TRUE)		
	A	B	C	D	E	F
1	Price_euros	Ram				
2	369	4				
3	400	4				
4	498.9	4		PRICE Lookup	RAM (GB)	
5	699	8		1300	=VLOOKUP(D5,A1:B10,2,TRUE)	
6	770	8				
7	860	8				
8	1158.7	8				
9	1339.69	8				
10	1495	16				
11						

- After entering the formula, press enter. The value returned, in this case, is 8GB for the price of \$1300.

E5					=VLOOKUP(D5,A1:B10,2,TRUE)
	A	B	C	D	E
1	Price_euros	Ram			
2	369	4			
3	400	4			
4	498.9	4		PRICE Lookup	RAM (GB)
5	699	8		1300	8
6	770	8			
7	860	8			
8	1158.7	8			
9	1339.69	8			
10	1495	16			
11					

How to Use VLOOKUP for Multiple Criteria?

With a helper column, you can give multiple criteria that the VLOOKUP genetically cannot handle. In our example, we will be calculating the price based on both company and product.

For example, if we have to look up the price of an Apple MacBook Pro, where the company name and product name are in two different columns, we use a helper column. This column will store the concatenated values of both columns.

Follow the steps below to perform VLOOKUP with multiple criteria.

- First, right-click on a column header and click on Insert. This will help you insert a column to the left of the Company column. Name it as 'Company & Product'.

	A	B	C	D	E	F
1	ID	Company		TypeName	Price_euros	Ram
2	1	Apple		Ultrabook	1339.69	8
3	2	Lenovo		Notebook	369	4
4	3	HP		Notebook	860	8
5	4	Microsoft		Ultrabook	1089	4
6	5	Dell		Notebook	498.9	4
7	6	Acer		Notebook	400	4
8	7	Apple		Ultrabook	1158.7	8
9	8	Asus		Ultrabook	1495	16
10	9	Acer		Ultrabook	770	8
11	10	HP		Ultrabook	699	8
12						
13						
14						

- On creating the helper column, enter the formula =C2&"-"&D2. Then, drag the formula down to the rest of the cells in the column.

SUM		X		✓		fx		=C2&"-"&D2	
	A	B	C	D	E	F	G	H	
1	ID	Company&Product	Company	Product	TypeName	Price_euros	Ram		
2	1	=C2&"-"&D2	Apple	MacBook Pro	Ultrabook	1339.69	8		
3	2		Lenovo	IdeaPad 320-15ISK	Notebook	369	4		
4	3		HP	ProBook 440	Notebook	860	8		
5	4		Microsoft	Surface Laptop	Ultrabook	1089	4		
6	5		Dell	Inspiron 3567	Notebook	498.9	4		
7	6		Acer	Aspire 3	Notebook	400	4		
8	7		Apple	Macbook Air	Ultrabook	1158.7	8		
9	8		Asus	ZenBook	Ultrabook	1495	16		
10	9		Acer	Swift 3	Ultrabook	770	8		
11	10		HP	Pavilion 15	Ultrabook	699	8		
12									

- On creating the concatenated column successfully, we can now look for the value. Here, we look for the price of an Apple MacBook Pro.
- So, we enter the formula =VLOOKUP(B15&"-"&C15,B1:G11,5,FALSE)

	A	B	C	D	E	F	G	H
1	ID	Company&Product	Company	Product	TypeName	Price_euros	Ram	
2	1	Apple-MacBook Pro	Apple	MacBook Pro	Ultrabook	1339.69	8	
3	2	Lenovo-IdeaPad 320-15ISK	Lenovo	IdeaPad 320-15ISK	Notebook	369	4	
4	3	HP-ProBook 440	HP	ProBook 440	Notebook	860	8	
5	4	Microsoft-Surface Laptop	Microsoft	Surface Laptop	Ultrabook	1089	4	
6	5	Dell-Inspiron 3567	Dell	Inspiron 3567	Notebook	498.9	4	
7	6	Acer-Aspire 3	Acer	Aspire 3	Notebook	400	4	
8	7	Apple-Macbook Air	Apple	Macbook Air	Ultrabook	1158.7	8	
9	8	Asus-ZenBook	Asus	ZenBook	Ultrabook	1495	16	
10	9	Acer-Swift 3	Acer	Swift 3	Ultrabook	770	8	
11	10	HP-Pavilion 15	HP	Pavilion 15	Ultrabook	699	8	
12								
13								
14		Company	Product	Price				
15		Apple	MacBook Pro	=VLOOKUP(B15&"-"&C15,B1:G11,5,FALSE)				
16								

- The first parameter helps to concatenate the lookup value, i.e., B15&"-" &C15. We can also use the concatenate formula, i.e., CONCATENATE(B15,"-",C15). The second parameter is the table range. The third parameter specifies the column index, which returns a value from that column. The final parameter is FALSE, as we are looking for an Exact Match. On pressing enter, the price will be returned as follows.

	A	B	C	D	E	F	G
1	ID	Company&Product	Company	Product	TypeName	Price_euros	Ram
2	1	Apple-MacBook Pro	Apple	MacBook Pro	Ultrabook	1339.69	8
3	2	Lenovo-IdeaPad 320-15ISK	Lenovo	IdeaPad 320-15ISK	Notebook	369	4
4	3	HP-ProBook 440	HP	ProBook 440	Notebook	860	8
5	4	Microsoft-Surface Laptop	Microsoft	Surface Laptop	Ultrabook	1089	4
6	5	Dell-Inspiron 3567	Dell	Inspiron 3567	Notebook	498.9	4
7	6	Acer-Aspire 3	Acer	Aspire 3	Notebook	400	4
8	7	Apple-Macbook Air	Apple	Macbook Air	Ultrabook	1158.7	8
9	8	Asus-ZenBook	Asus	ZenBook	Ultrabook	1495	16
10	9	Acer-Swift 3	Acer	Swift 3	Ultrabook	770	8
11	10	HP-Pavilion 15	HP	Pavilion 15	Ultrabook	699	8
12							
13							
14		Company	Product	Price			
15		Apple	MacBook Pro	1339.69			

Frequency

=FREQUENCY(data_array, bins_array)

The FREQUENCY function uses the following arguments:

1. **Data_arrays** (It is a required argument) – This is an array or reference to a set of values for which you want to count frequencies.
2. **Bins_array** (It is a required argument) – This is an array of intervals (“bins”) for grouping values.

Remember that:

1. If data_array contains no values, FREQUENCY returns an array of zeros.
2. If bins_array contains no values, FREQUENCY returns the number of elements in data_array.

To create a frequency distribution using FREQUENCY:

- We need to enter numbers that represent the bins we want to group values into.
- Make a selection the same size as the range that contains bins, or greater by one, if we want to include the extra item.
- Enter the FREQUENCY function as an array formula using Control+Shift+Enter.

How to use the FREQUENCY Function in Excel?

As a worksheet function, FREQUENCY can be entered as part of a formula in a cell of a worksheet. To understand the uses of the function, let us consider a few examples:

Example 1

Suppose we are a toy manufacturing company. We can use the FREQUENCY function to count the number of children falling into three different age ranges, which are specified by the bins_array (stored in cells B2-B3 of the spreadsheet).

	A	B	C	D
1				
2		FREQUENCY Function		
3				
4		Age	Bins	
5		5	4.00	
6		8	4.00	
7		11	7.00	
8		9	7.00	
9		5	4.00	
10		7	4.00	
11		6	4.00	
12		3	6.00	
13		4	6.00	
14		2	6.00	
15		1	6.00	
16		2	6.00	
17		5	4.00	

In the table above, the bins_array values specify the maximum values for the age ranges. Therefore, in this example, the ages are to be split into the ranges 0-4 years, 5-8 years and 9 years+.

We will enter the formula below:

IRR					
X ✓ f_x					
=FREQUENCY(B5:B17,C5:C17)					
	A	B	C	D	E
1					
2		FREQUENCY Function			
3					
4		Age	Bins	Frequency	
5		5	4.00	=FREQUENCY(B5:B17,C5:C17)	
6		8	4.00	1.20	
7		11	7.00	2.10	
8		9	7.00	2.10	
9		5	4.00	1.20	
10		7	4.00	1.20	
11		6	4.00	1.20	
12		3	6.00	1.80	
13		4	6.00	1.80	
14		2	6.00	1.80	
15		1	6.00	1.80	
16		2	6.00	1.80	
17		5	4.00	1.92	

FREQUENCY would be entered as an array formula after we select a range of adjacent cells into which we want the returned distribution to appear.

We used CTRL+SHIFT+ENTER to get curly brackets for array formulas. The results we get are shown below:

D5							=FREQUENCY(B5:B17,C5:C17)
	A	B	C	D	E	F	
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							
11							
12							
13							
14							
15							
16							
17							

FREQUENCY Function

Age	Bins	Frequency
5	4.00	5.00
8	4.00	1.20
11	7.00	2.10
9	7.00	2.10
5	4.00	1.20
7	4.00	1.20
6	4.00	1.20
3	6.00	1.80
4	6.00	1.80
2	6.00	1.80
1	6.00	1.80
2	6.00	1.80
5	4.00	1.92

Example 2

We can use the FREQUENCY function to count unique values in a range with some criteria. Suppose we are given a list of employees who participated in an activity, along with the time expended on the activity.

Looking at the data below, we can see that the same employee names appear more than once, so what we want is a count of unique names.

B2							FREQUENCY Function
	A	B	C	D	E		
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							
11							
12							
13							

FREQUENCY Function

Name	Activity	Hours
Jimmy	Mentoring	4
Sean	In house training	2
William	Mentoring	3
Linda	Mentoring	5
William	In house training	6
Jimmy	Mentoring	4
Sean	In house training	2
Sean	Mentoring	5
William	In house training	6

The formula we will use is:

=SUM(--(FREQUENCY(IF(B2:B10=F1,MATCH(A2:A10,A2:A10,0)),ROW(A2:A10)-ROW(A2)+1)>0))--:

IRR

=SUM(--(FREQUENCY(IF(C5:C13=C15,MATCH(B5:B13,B5:B13,0)),ROW(B5:B13)-ROW(B5)+1)>0))

Name	Activity	Hours
Jimmy	Mentoring	4
Sean	In house training	2
William	Mentoring	3
Linda	Mentoring	5
William	In house training	6
Jimmy	Mentoring	4
Sean	In house training	2
Sean	Mentoring	5
William	In house training	6

Mentoring

=SUM(--(FREQUENCY(IF(C5:C13=C15,MATCH(B5:B13,B5:B13,0)),ROW(B5:B13)-ROW(B5)+1)>0))

Use CTRL+SHIFT+ENTER to get curly brackets for array formulas. We will get the result below:

D15

{=SUM(--(FREQUENCY(IF(C5:C13=C15,MATCH(B5:B13,B5:B13,0)),ROW(B5:B13)-ROW(B5)+1)>0))}

Name	Activity	Hours
Jimmy	Mentoring	4
Sean	In house training	2
William	Mentoring	3
Linda	Mentoring	5
William	In house training	6
Jimmy	Mentoring	4
Sean	In house training	2
Sean	Mentoring	5
William	In house training	6

Mentoring

4

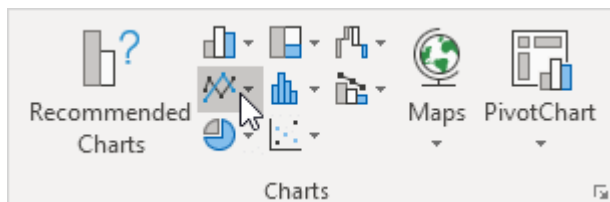
Create a Chart

To create a line chart, execute the following steps.

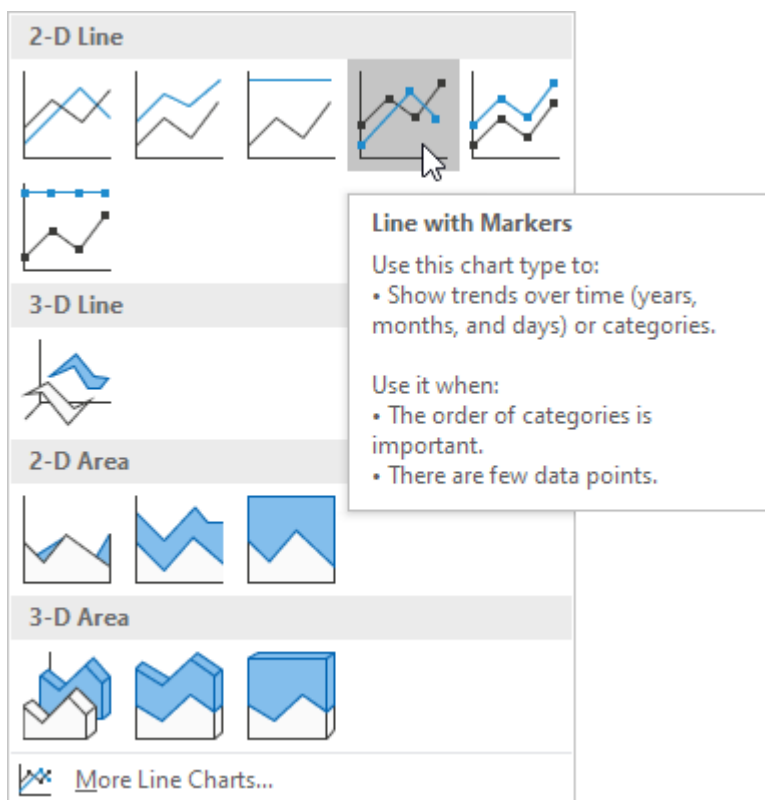
1. Select the range A1:D7.

	A	B	C	D	E
1	Month	Bears	Dolphins	Whales	
2	Jan	8	150	80	
3	Feb	54	77	54	
4	Mar	93	32	100	
5	Apr	116	11	76	
6	May	137	6	93	
7	Jun	184	1	72	
8					

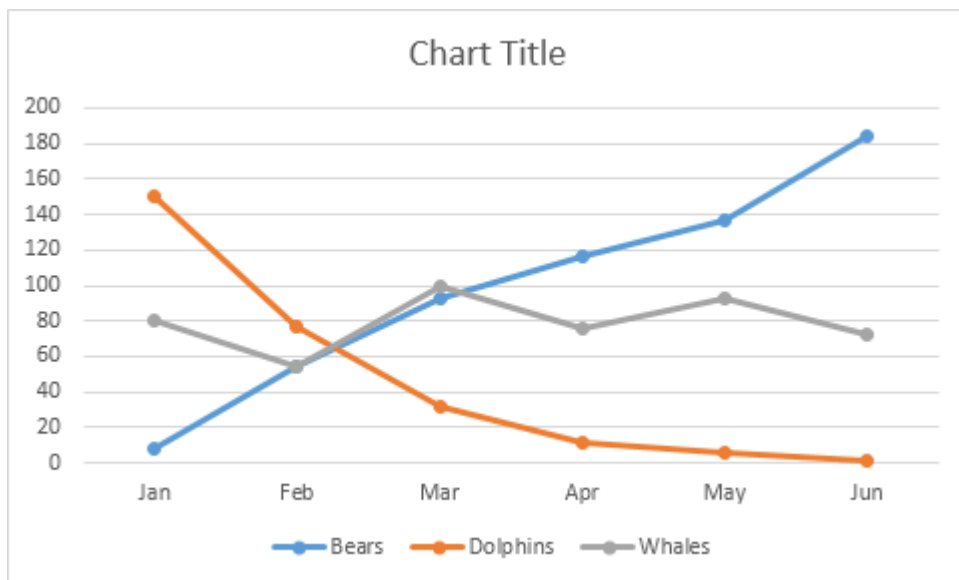
2. On the Insert tab, in the Charts group, click the Line symbol.



3. Click Line with Markers.



Result:

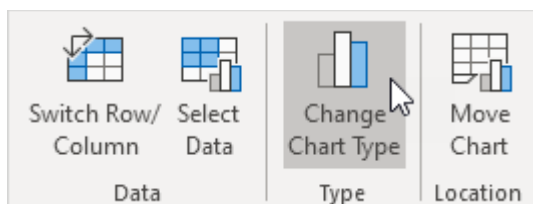


Note: enter a title by clicking on Chart Title. For example, Wildlife Population.

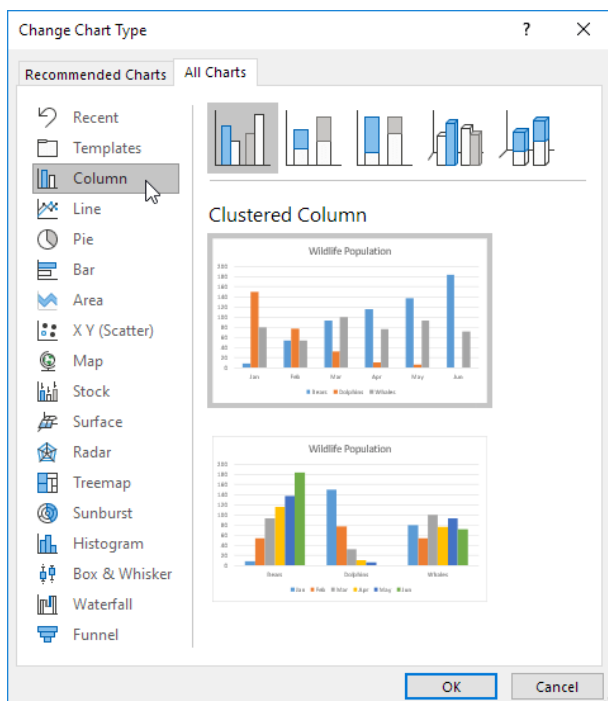
Change Chart Type

You can easily change to a different type of chart at any time.

1. Select the chart.
2. On the Chart Design tab, in the Type group, click Change Chart Type.

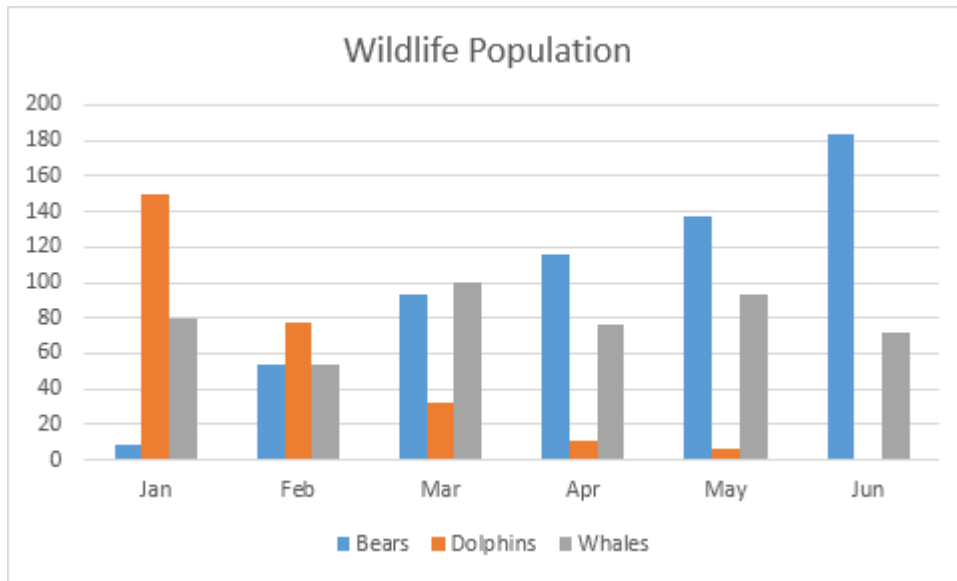


3. On the left side, click Column.



4. Click OK.

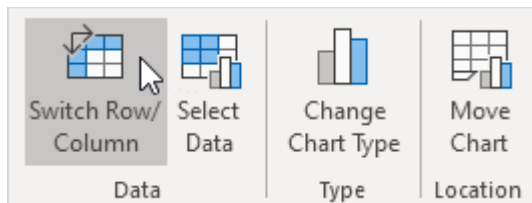
Result:



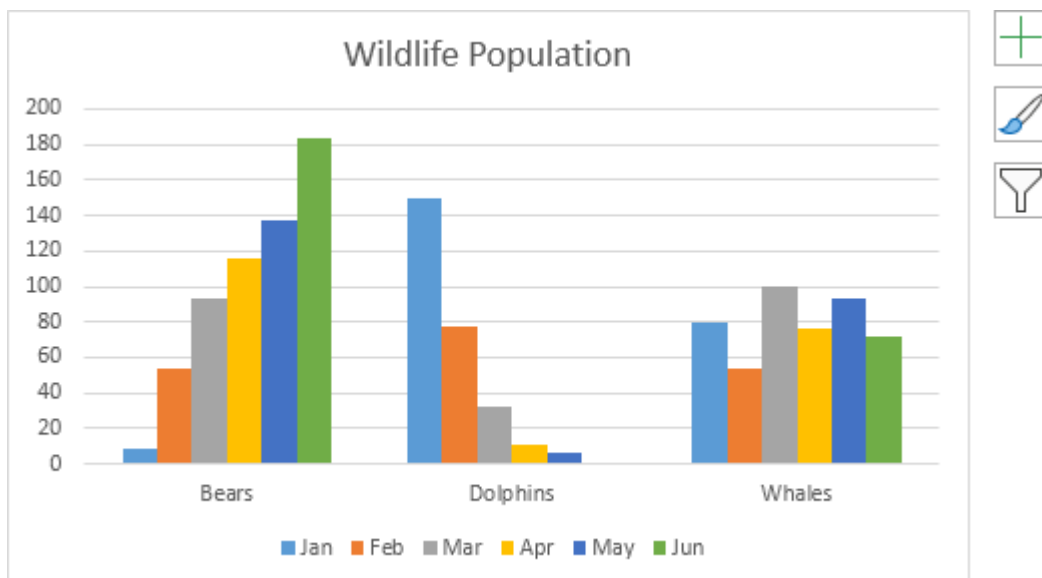
Switch Row/Column

If you want to display the animals (instead of the months) on the horizontal axis, execute the following steps.

1. Select the chart.
2. On the Chart Design tab, in the Data group, click Switch Row/Column.



Result:

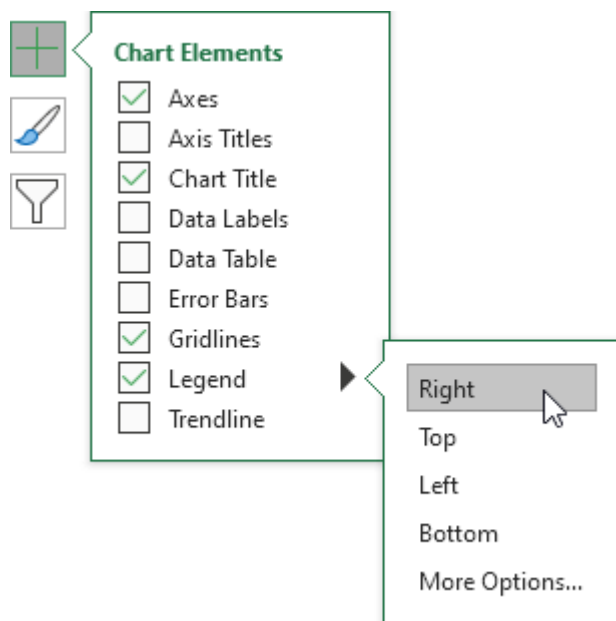


Legend Position

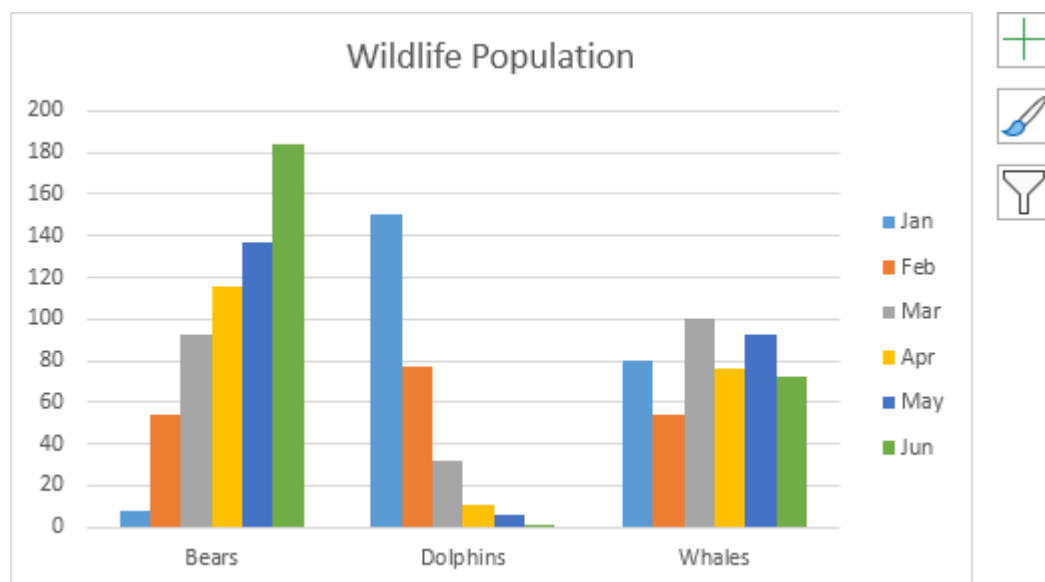
To move the legend to the right side of the chart, execute the following steps.

1. Select the chart.

2. Click the + button on the right side of the chart, click the arrow next to Legend and click Right.



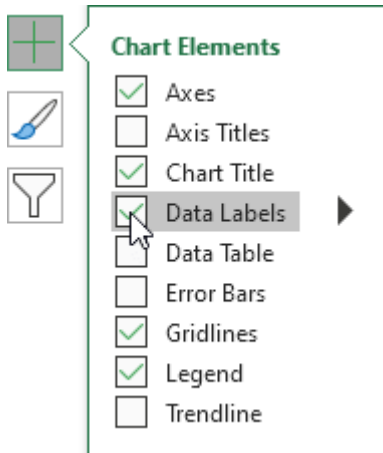
Result:



Data Labels

You can use data labels to focus your readers' attention on a single data series or data point.

1. Select the chart.
2. Click a green bar to select the Jun data series.
3. Hold down CTRL and use your arrow keys to select the population of Dolphins in June (tiny green bar).
4. Click the + button on the right side of the chart and click the check box next to Data Labels.



Result:

